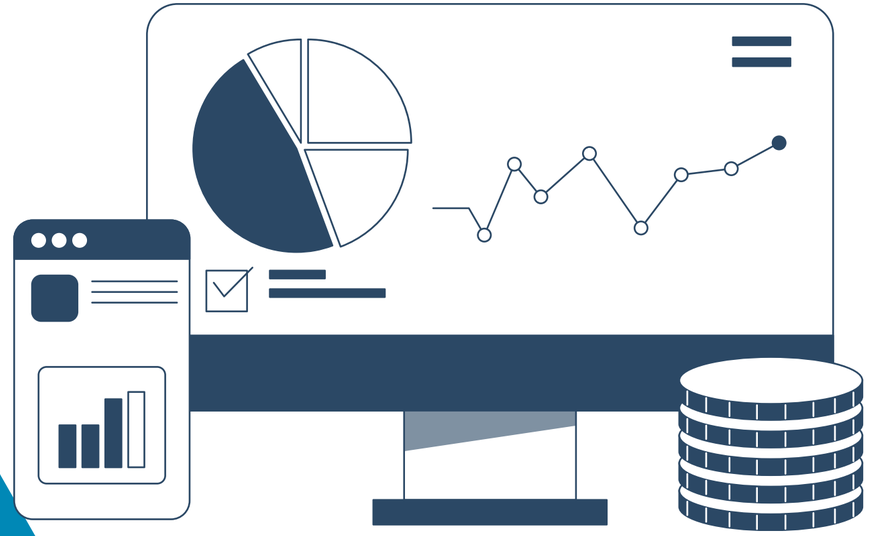
DATA ANALYTICS

30 days



DAY 23

Agenda

- What is Pandas?
- Understanding Series and DataFrame
- Loading Data from a CSV File
- Exploring and Understanding a Dataset
- Checking for Missing Values
- Extracting Business Insights from Data

What is Pandas?

Pandas is a Python library used for working with structured data.

Why Use Pandas?

- Read data from CSV, Excel, Database
- Clean messy data
- Analyze data quickly
- Generate business insights

Understanding Series

A Series is a single column of data.

```
import pandas as pd


sales = pd.Series([100, 200, 300, 400])
print(sales)
```

Understanding DataFrame

A DataFrame is a collection of rows and columns.

```
import pandas as pd
data = {
    'Product':['Laptop','Phone','Mouse'],
    'Price':[50000,20000,500]
}
df = pd.DataFrame(data)
print(df)
```

Loading Data from a CSV File

CSV is Comma Separated Values

```
import pandas as pd


df = pd.read_csv('sales.csv')
```

Product,Price

Laptop,50000

Phone,20000

Mouse,500

Complete Example

- import pandas as pd
- df = pd.read_csv('sales.csv')
- print(df.head())
- print(df.shape)
- print(df.columns)
- print(df.info())
- print(df.describe())
- print(df.isnull().sum())
- print("Total Orders:", df.shape[0])
- print("Average Revenue:", df['Revenue'].mean())

Operation on Dataset

Viewing Data with Head()

```
df.head()
```

Understanding Dataset Size

```
df.shape
```

Viewing Data with Tail()

```
df.tail()
```

Viewing Column Names

```
df.columns
```

Operation on Dataset

Dataset Information

```
df.info()
```

Statistical Summary

```
df.describe()
```

Data Types

```
df.dtypes
```

Find Missing Values

```
df.isnull().sum()
```

THANK YOU

FOR YOUR ATTENTION